Organic Chemistry Master Reaction Roadmap

Class 12 CBSE & JEE/NEET Revision Sheet

Chapter / Focus Area: Haloalkanes, Alcohols, Aldehydes, Ketones & Amines

I. Alkane & Alkene Functional Group Interconversions

1. Alkene + HBr (in presence of Peroxide) -> Anti-Markovnikov 1-Bromoalkane
2. Alkene + HBr (no peroxide) -> Markovnikov 2-Bromoalkane (Carbocation intermediate)
3. Alkene + cold dilute alkaline $KMnO_4$ (Baeyer's Reagent) -> Vicinal Diols

II. Oxidation Roadmaps (Alcohols to Carboxylic Acids)

1. Primary Alcohol (R-CH_2-OH) + PCC / CrO_3 -> Aldehyde (R-CHO)
2. Primary Alcohol + Alkaline $KMnO_4$ / Acidified $K_2Cr_2O_7$ -> Carboxylic Acid (R-COOH)
3. Secondary Alcohol (R-CH(OH)-R') + CrO_3 -> Ketone (R-CO-R')
4. Tertiary Alcohol + Cu / 573 K -> Dehydration to Alkene (2-Methylpropene)

III. Key Named Reaction Mechanism Summary

• Aldol Condensation: Requires alpha-hydrogen, forms beta-hydroxyaldehyde.

• Cannizzaro Reaction: Formaldehyde / Benzaldehyde (no alpha-H) + 50% NaOH -> Alcohol + Carboxylate.

• Reimer-Tiemann Reaction: Phenol + $CHCl_3$ + aq. NaOH -> Salicylaldehyde.

• Kolbe's Reaction: Phenol + NaOH + CO_2 (400 K, 4-7 atm) -> Salicylic Acid.